

Patient Health Record Consent Management System

Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a patient to grant time-bounded access to specific medical records to a verified healthcare provider.	High	Pass: When a patient selects a record, verified provider, and expiry time, the system grants access only until the specified expiry time. Fail: Access is granted without a valid provider, record selection, or expiry time.	Core functionality that allows patients to control access to their health records.
FR-002	Functional	The system shall allow a patient to revoke an active consent permission before its expiry time	High	Pass: When the patient revokes an active consent, the selected provider can no longer access the permitted record immediately. Fail: The provider can still access the record after revocation.	Gives patients direct control over previously granted permissions.
FR-003	Functional	The system shall allow a patient to view all active and expired consent permissions associated with their health records.	Medium	Pass: The system displays each consent with the provider name, permitted record, consent status, and expiry time. Fail: Any active or expired consent is missing or displayed with incorrect information.	Helps patients monitor who has or had access to their medical information.
FR-004	Functional	The system shall automatically revoke access when the consent expiry time is reached.	High	Pass: A provider attempting to access a record after the expiry timestamp is denied access. Fail: The provider successfully accesses the record after consent expiration.	Ensures that time-bound consent is enforced automatically.
FR-005	Functional	The system shall allow a verified healthcare provider to access only the specific patient records for which valid consent has been granted.	High	Pass: A provider with valid consent can access only the authorised records, while requests for unauthorised or expired records are denied. Fail: A provider accesses a record without valid consent.	Protects patient privacy through granular access control.
NFR-001	Non-Functional & Auditability	The system shall permanently record all consent grant, revoke, expiry, and healthcare-provider access events in an append-only audit trail.	High	Pass: Each tested event creates an audit entry containing the event type, timestamp, patient, provider, and affected record, and existing entries cannot be modified or deleted through normal system operations. Fail: Any event is missing from the	Provides accountability and traceability for sensitive health-data access.

				audit trail or an existing entry can be altered.	
NFR-002	Non-Functional & Performance	The system shall return the result of a consent validation or authorised medical-record access request within 3 seconds under normal operating conditions.	Medium	Pass: Performance testing shows that at least 95% of tested requests complete within 3 seconds. Fail: More than 5% of tested requests exceed 3 seconds.	Ensures that authorised healthcare providers can access required information without unnecessary delay.